

L8b: Utility-Based Allocation and Bets on the Future

CHEME 5660 Cornell University

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Objectives for Today

By the end of this lecture, you should be able to:

- Solve budget-constrained utility problems: find the Cobb–Douglas share counts in closed form with SIM preference weights, and state how the CES elasticity moves the allocation.
- Choose a point on the capital allocation line: derive the utility-maximizing risk-free fraction and infer a client's risk aversion.
- Describe a bet on a future condition: write it as an abstract asset, compute its expected NPV, and explain why buyers with different beliefs value it differently.

Lectures and Examples

Lecture: Utility-Based Allocation and Bets on the Future

Example 1: Build a Maximum-Utility Portfolio Allocator
Which firms, and how much of each? How does re-solving through 2025 compare with buy-and-hold?

Interview: Measure Risk Aversion with an Income-Gamble Interview
Would you take a job that could double your income or cut it by a third?

Example 2: Choose the Optimal Point on the Capital Allocation Line
Which risk-free fraction maximizes utility, and what A does each answer imply?

Both examples use the default firms unless you uncomment the ticker-file lines.

Concept Review: Utility Functions

Utility functions describe a decision-maker's preferences over choices, outcomes, or states of the world. A larger utility is always better.

An agent chooses from alternatives $\mathcal{X} = \{x_1, \ldots, x_n\}$. A utility function $U : \mathcal{X} \to \mathbb{R}$ assigns each one a number, in *utils*, that ranks the agent's preferences:

$$U(x_i) > U(x_j) \iff x_i \succ x_j \qquad \text{and} \qquad U(x_i) = U(x_j) \iff x_i \sim x_j$$

- Ordinal: U ranks alternatives without measuring how much better one is, so any order-preserving transformation (e.g., $\ln U$ for $U > 0$) keeps the same ranking.
- Key idea: a rational decision-maker always chooses the alternative with the largest utility.

Common Utility Functions

Utility functions, and their parameters, are subjective. Five common forms for a bundle of goods $\mathbf{x} = (x_1, \dots, x_n)$ with $x_i \geq 0$ and $\alpha_i > 0$:

Utility function	Form	Models
Linear	$U(\mathbf{x}) = \sum_i \alpha_i x_i$	goods that are perfect substitutes
Logarithmic ($\beta > 0$)	$U(\mathbf{x}) = \ln(\sum_i \alpha_i x_i + \beta)$	diminishing satisfaction from more goods
Cobb–Douglas	$U(\mathbf{x}) = \prod_i x_i^{\alpha_i}$	goods held together
Leontief	$U(\mathbf{x}) = \min(x_1/\alpha_1, \dots, x_n/\alpha_n)$	perfect complements, fixed proportions
CES ($\eta > 0$, $\eta \neq 1$)	$U(\mathbf{x}) = \left(\sum_i \alpha_i x_i^{(\eta-1)/\eta}\right)^{\eta/(\eta-1)}$	substitution set by the elasticity η

We use the Cobb–Douglas and CES utilities next. As η changes, CES moves between the linear ($\eta \to \infty$), Cobb–Douglas ($\eta \to 1$, with $\sum_i \alpha_i = 1$), and Leontief ($\eta \to 0$) forms.

Cobb–Douglas Utility over Holdings

An investor chooses share counts to maximize a holding utility, which has no variance term. Risk enters through the preference weights $\gamma_i \in (-1, 1)$. With them as exponents, the Cobb–Douglas problem is:

$$\begin{aligned}
&\underset{n_1,\dots,n_{|\mathcal{P}|}}{\text{maximize}} && U(n_1, \dots, n_{|\mathcal{P}|}) = \prod_{i \in \mathcal{P}} n_i^{\gamma_i} && \\
&\text{subject to} && \sum_{i \in \mathcal{P}} n_i \, S_i(t) + \text{cash}(t) = W_{\mathcal{P}}(t) && \text{(budget constraint)} \\
& && n_i \geq n_{\min} \quad \forall \, i \in \mathcal{P}, \qquad \text{cash}(t) \geq 0 && \text{(share floor, no borrowing)}
\end{aligned}$$

where $\mathcal{P}$ is the asset set, n_i the share count of asset i, $S_i(t)$ its price (USD per share), $W_{\mathcal{P}}(t)$ the wealth, and $\text{cash}(t)$ the uninvested wealth (USD). The floor is $n_{\min} > 0$ shares, and cash adds no utility.

The Cobb–Douglas Allocation

Assets with $\gamma_i > 0$ form the preferred set $\mathcal{A}^+$, the basket. The rest form $\mathcal{A}^- = \{i : \gamma_i \le 0\}$. The floor keeps U finite when $\gamma_i < 0$.

With positive prices, enough wealth to fund the floors, and a nonempty basket, non-preferred assets stay at their floors and cash is zero, which leaves the net budget:

$$W_{\text{adj}} = W_{\mathcal{P}}(t) - n_{\min} \sum_{k \in \mathcal{A}^-} S_k(t)$$

When preferred holdings stay above their floors, the allocation is:

$$n_i^\star = \underbrace{\frac{\gamma_i}{\sum_{j \in \mathcal{A}^+} \gamma_j}}_{\text{preference share}} \cdot \underbrace{\frac{W_{\text{adj}}}{S_i(t)}}_{\text{budget in shares}} \qquad (i \in \mathcal{A}^+), \qquad n_k^\star = n_{\min} \quad (k \in \mathcal{A}^-)$$

Derivation: Cobb--Douglas companion, including what to do when a preferred holding reaches its floor.

CES Utility: A Concentration Dial

CES adds the elasticity $\eta > 0$, which sets how willingly the allocator trades one preferred asset for another. For $\eta \neq 1$:

$$U_{\mathrm{CES}} = \Big(\sum_{i \in \mathcal{A}^+} \gamma_i \, n_i^{(\eta-1)/\eta} \Big)^{\eta/(\eta-1)}$$

$$n_i^\star = W_{\mathrm{adj}} \, \frac{(\gamma_i / S_i(t))^\eta}{\sum_{j \in \mathcal{A}^+} S_j(t) (\gamma_j / S_j(t))^\eta} \quad (i \in \mathcal{A}^+)$$

Share counts above the floors are proportional to the *bang for the buck* $\gamma_i / S_i(t)$ raised to η. Three limits:

- Unit elasticity ($\eta \to 1$): the Cobb–Douglas allocation.
- Large elasticity ($\eta \to \infty$): the unique largest $\gamma_i / S_i(t)$ gets the whole budget above the floors. Every other asset falls to its floor.
- Small elasticity ($\eta \to 0^+$): equal share counts of every preferred asset.

Proofs: CES companion, including why CES dollar weights depend on share prices when $\eta \neq 1$.

Preference Weights from the Single Index Model

Each trading day, we compute $\gamma_i(t)$ from the SIM parameters (α_i, β_i) and two SPY inputs: the recent market growth rate $\tilde{g}_{M,t}$ and the market-state signal ξ_t, which says whether the market has been rising or falling:

$$\gamma_i(t) = \tanh\left(h \left[\frac{\alpha_i}{\beta_i^{\xi_t}} + \beta_i^{1-\xi_t}\, \tilde{g}_{M,t} \right] \right)$$

Here $\tilde{g}_{i,t} = \alpha_i + \beta_i \tilde{g}_{M,t}$ is the SIM's expected growth today, $h = 1$ year makes the argument dimensionless, and the model assumes $\beta_i > 0$.

- Sign picks the basket: $\gamma_i(t) > 0$ when $\tilde{g}_{M,t} > -\alpha_i/\beta_i$. As $\tilde{g}_{M,t}$ rises, the basket grows. As it falls, only the lowest-threshold assets stay. With an empty basket, the allocator holds the floors plus cash.
- Magnitude splits the budget: bearish ($\xi_t > 0$) tilts it toward low-beta assets, bullish toward high-beta.

Market Inputs from SPY

Both inputs come from SPY prices through exponential moving averages (EMAs), which weight recent prices most. For windows of L trading days:

$$\bar{S}_t = \omega\, S_t + (1-\omega)\, \bar{S}_{t-1}, \quad \omega = \frac{2}{L+1}$$

$$\boxed{\xi_t = -G\left(\frac{\bar{S}_t^{\text{short}}}{\bar{S}_t^{\text{long}}} - 1\right)}$$

With $L_{\text{short}} = 21$ and $L_{\text{long}} = 63$, ξ_t compares recent prices with the quarter's average:

- $\bar{S}_t^{\text{short}} > \bar{S}_t^{\text{long}}$: $\xi_t < 0$, bullish. The market has been rising.
- $\bar{S}_t^{\text{short}} = \bar{S}_t^{\text{long}}$: $\xi_t = 0$, neutral, no tilt.
- $\bar{S}_t^{\text{short}} < \bar{S}_t^{\text{long}}$: $\xi_t > 0$, bearish. The market has been falling.

We choose the gain $G > 0$ so that $|\xi_t| \leq 1$ on 99% of training days, and $\tilde{g}_{M,t}$ is the 252-day EMA of $g_{M,t} = \ln(S_t/S_{t-1})/\Delta t$, with $\Delta t = 1/252$ year. Both inputs use prices through day $t-1$.

Example 1: Build a Maximum-Utility Portfolio Allocator

The Complete Portfolio

As in L7a, a complete portfolio c puts a fraction w_f of wealth in the risk-free asset and $1 - w_f$ in the tangent portfolio $\mathcal{T}$. Its expected growth rate and variance are:

$$\mathbb{E}[g_c] = g_f + (1 - w_f)(\mathbb{E}[g_\mathcal{T}] - g_f) \qquad \text{expectation is linear}$$

$$\sigma^2_{g,c} = (1 - w_f)^2 \, \sigma^2_{g,\mathcal{T}} \qquad \text{the risk-free asset adds no variance}$$

where growth rates are in inverse years and g_f is the risk-free growth rate. A negative w_f means borrowing at g_f to buy more of $\mathcal{T}$.

Which w_f should a given investor choose? The answer depends on how risk averse the investor is.

Measuring Risk Aversion

The certainty equivalent (CE) is the sure wealth whose utility equals the expected utility of an uncertain wealth W. The risk premium is the gap:

$$\underbrace{U(\mathrm{CE})}_{\text{utility of the sure wealth}} = \underbrace{\mathbb{E}[U(W)]}_{\text{expected utility}} \qquad \mathrm{RP} = \underbrace{\mathbb{E}[W]}_{\text{expected wealth}} - \underbrace{\mathrm{CE}}_{\text{sure wealth}}$$

The more concave U, the further CE falls below $\mathbb{E}[W]$. Arrow and Pratt (1964) measure this curvature with the relative risk aversion $\bar{r}$. Over one trading day, today's wealth W_0 grows to $W = W_0 e^{g_c \Delta t}$:

$$\bar{r} = -W_0 \frac{U''(W_0)}{U'(W_0)} \qquad \text{relative risk aversion}$$

$$\mathrm{RP} \approx -\frac{1}{2} \frac{U''(W_0)}{U'(W_0)} \, \mathrm{Var}(W) \qquad \text{Pratt (1964), small risk}$$

$$\mathrm{Var}(W) \approx W_0^2 \, (\Delta t)^2 \sigma_{g,c}^2 \qquad W \approx W_0(1 + g_c \Delta t) \text{ for one day}$$

$$\frac{\mathrm{RP}}{W_0} \approx \frac{\bar{r}}{2} (\Delta t)^2 \sigma_{g,c}^2 \qquad \text{substitute, divide by } W_0$$

The Certainty-Equivalent Growth Rate

The certainty-equivalent growth rate g_{CE} is the sure rate that replaces the risky g_c:

$$\mathrm{CE} = W_0\, e^{g_{\mathrm{CE}}\Delta t} \qquad \text{definition of } g_{\mathrm{CE}}$$

$$\frac{\mathbb{E}[W]}{W_0} \approx 1 + \mathbb{E}[g_c]\,\Delta t + \tfrac{1}{2}(\Delta t)^2\sigma_{g,c}^2 \qquad \text{expand } e^x,\ \text{take expectations},\ \mathbb{E}[g_c]^2 \ll \sigma_{g,c}^2$$

$$\frac{\mathrm{CE}}{W_0} = \frac{\mathbb{E}[W]}{W_0} - \frac{\mathrm{RP}}{W_0} \qquad \mathrm{RP} = \mathbb{E}[W] - \mathrm{CE}$$

$$\approx 1 + \mathbb{E}[g_c]\,\Delta t - \frac{\bar{r}-1}{2}\,(\Delta t)^2\sigma_{g,c}^2 \qquad \text{substitute, collect}$$

$$g_{\mathrm{CE}} \approx \mathbb{E}[g_c] - \frac{\bar{r}-1}{2}\,\Delta t\,\sigma_{g,c}^2 \qquad \ln(1+x) \approx x,\ \text{divide by } \Delta t$$

The variance term lowers g_{CE} only when $\bar{r} > 1$, which we assume from here on.

Every step: lecture notebook, Measuring risk aversion.

Mean-Variance Utility

The investor wants the largest sure growth rate. With $A \equiv \bar{r} - 1$, g_{CE} is the mean-variance utility, which we maximize over w_f:

$$\boxed{\underset{w_f}{\text{maximize}} \quad U(w_f) = \mathbb{E}[g_c] - \frac{A}{2}\,\Delta t\,\sigma_{g,c}^2 \approx g_{\mathrm{CE}}}$$

The risk-aversion coefficient $A > 0$ (dimensionless) sets how much the investor dislikes variance. We assume $\mathbb{E}[g_{\mathcal{T}}] > g_f$. On the line, both terms depend only on w_f:

$$U(w_f) = g_f + (1 - w_f)(\mathbb{E}[g_{\mathcal{T}}] - g_f) - \tfrac{A}{2}(1 - w_f)^2\,\Delta t\,\sigma_{g,\mathcal{T}}^2 \qquad \text{substitute the line}$$

$$\frac{dU}{dw_f} = -(\mathbb{E}[g_{\mathcal{T}}] - g_f) + A\,(1 - w_f)\,\Delta t\,\sigma_{g,\mathcal{T}}^2 \qquad \text{differentiate}$$

$$0 = -(\mathbb{E}[g_{\mathcal{T}}] - g_f) + A\,(1 - w_f^\star)\,\Delta t\,\sigma_{g,\mathcal{T}}^2 \qquad \text{first-order condition}$$

The Optimal Complete Portfolio

Solving the first-order condition for the fraction in $\mathcal{T}$ gives:

$$\boxed{1 - w_f^\star = \frac{\mathbb{E}[g_{\mathcal{T}}] - g_f}{A\,\Delta t\,\sigma_{g,\mathcal{T}}^2}}$$

The second derivative, $-A\,\Delta t\,\sigma_{g,\mathcal{T}}^2$, is negative, so this is a maximum. The fraction in $\mathcal{T}$ rises with the excess growth and falls as A or $\sigma_{g,\mathcal{T}}^2$ grows.

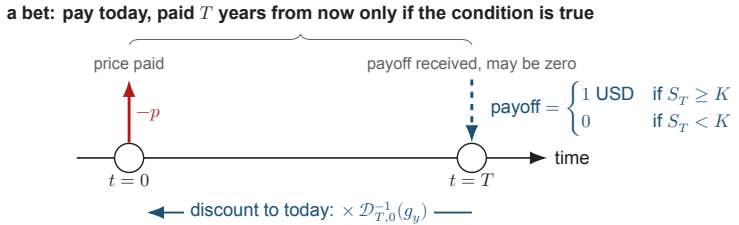
An indifference curve, $\mathbb{E}[g_c] = U + \frac{A}{2}\,\Delta t\,\sigma_{g,c}^2$, joins the portfolios with the same utility U. The optimum is where the highest reachable curve touches the line.

Solving the box for A gives the risk aversion behind a client's chosen $w_f < 1$.

Interview and Example 2: Income-Gamble Interview Choose the Optimal Point on the Capital Allocation Line

Derivatives: A Bet on the Future

Suppose we pay p today for a contract that pays 1 USD in T years if $S_T \geq K$ (the L5a trade rule), and nothing otherwise.



Discounting the payoff at the benchmark growth rate g_y gives a random NPV:

$$\text{NPV}_0 = -p + \mathcal{D}_{T,0}^{-1}(g_y)\, V(S_T) = \begin{cases} -p + e^{-g_y T} & \text{if } S_T \geq K \text{ (true)} \\ -p & \text{if } S_T < K \text{ (false)} \end{cases}$$

Expected NPV of the Bet

Under our beliefs about S_T, the expected NPV is:

$$\mathbb{E}[\text{NPV}_0] = (-p + e^{-g_y T})\underbrace{\mathbb{P}(S_T \geq K)}_{\text{true}} + (-p)\underbrace{\mathbb{P}(S_T < K)}_{\text{false}} \qquad \text{weight by probability}$$

$$= -p\underbrace{[\mathbb{P}(S_T \geq K) + \mathbb{P}(S_T < K)]}_{=1} + e^{-g_y T}\,\mathbb{P}(S_T \geq K) \qquad \text{collect the } p \text{ terms, probabilities sum to one}$$

$$= -p + e^{-g_y T}\underbrace{\mathbb{P}(S_T \geq K)}_{\text{true}}$$

The bet has a non-negative expected NPV when $p \leq e^{-g_y T}\,\mathbb{P}(S_T \geq K)$.

The bet is a derivative contract: its payoff depends on the future price of another asset. Every buyer pays one market price, yet buyers with different beliefs value the same contract differently.

How is the market price set?

Optional Advanced Material

The advanced index describes the optional derivation notebooks:

- Cobb–Douglas allocation with share floors: derive the allocation when a preferred holding reaches its floor, show why fixing it there and solving again gives the optimum, and work an example with two floor adjustments.
- CES limits and log-linear utility: derive the closed-form CES optimum and prove its three limits. Show that log-linear utility has the Cobb–Douglas optimizer, and why CES weights depend on share price levels.

Summary

- The single index model and Cobb–Douglas utility turn preferences into holdings: the signs of the preference weights pick the basket, and Cobb–Douglas splits the net budget in proportion to the positive weights. The CES elasticity moves the split toward concentration or equal share counts.
- Risk aversion picks the point on the capital allocation line: the certainty-equivalent growth rate is a mean-variance utility with $A = \bar{r} - 1$, and the risky fraction follows in closed form. A client's chosen fraction implies A.
- A bet on the future is an abstract asset with a random cash flow: its expected NPV depends on our beliefs, so buyers with different beliefs value it differently.

Utility makes investment decisions personal, but a derivative contract trades at one market price that every buyer pays.